

General Chemistry (CHEM 1310)

Midterm Examination 2

Name	GT ID (90...)
Section (A or B)	

Indicate your response to each question in the box or on the line(s) provided. For multiple-choice questions, fill in the circle next to your choice clearly and completely. For multiple-answer questions, fill in the circles completely next to statements you wish to select.

Academic Integrity Guidelines

1. This quiz is closed note, closed book, and closed Internet. You are permitted only a single crib sheet and a calculator lacking Internet connectivity.
2. You may not access any other resource or website aside from this document for the duration of the exam.
3. You may not collaborate with any other person (in person or electronically) on any portion of the exam, including general discussion of exam question topics prior to exam submission by all parties.
4. You are not permitted to screen-capture or otherwise record the questions on this exam.
5. All work submitted for this exam must be your own.

I affirm that I have abided by all academic integrity guidelines as listed above.

Signature _____

Do not open this exam until you are instructed to do so.

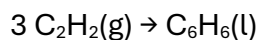
1. **5 points.** A sample of an ideal gas at 21.2 °C and 0.992 atm in a rigid 4.00-L container is heated to 41.8 °C. What is the pressure of the gas in atmospheres (atm) at the end of this process? The ideal gas constant is 8.314 J/K·mol or 0.08206 L·atm/K·mol.
- A. ☐ 0.932 atm
B. ☐ 1.06 atm
C. ☐ 1.96 atm
D. ☐ 0.503 atm
E. ☐ 1.12 atm
2. **6 points.** A gas cylinder with a volume of 50.0 L contains a mixture of He, Ar, and Kr gases at a total pressure of 30.0 atm. The partial pressures of the gases are as follows. Rank the gases from smallest (1) to greatest (3) number of moles in the mixture.

$$P(\text{He}) = 5.0 \text{ atm}$$

$$P(\text{Ar}) = 15.0 \text{ atm}$$

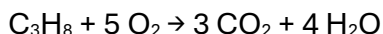
$$P(\text{Kr}) = 10.0 \text{ atm}$$

3. **5 points.** Gaseous acetylene (C_2H_2) can be converted to liquid benzene (C_6H_6) by certain catalysts according to the balanced chemical equation below.



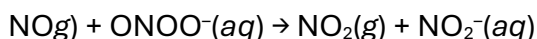
Acetylene at a temperature of 250. K in a sealed vessel with a volume of 0.600 L and initial pressure of 1.62 atm was subjected to this reaction. The pressure when the reaction was terminated was 0.34 atm. Assuming no change in the volume of the gas, what amount in moles (mol) of benzene was produced in this process? Enter your response in moles (mol) to the nearest 0.001 mol.

4. **5 points.** Which of the following statements is *true* regarding a 1.0 L sample of helium gas and a 1.0 L sample of argon gas at room temperature and atmospheric pressure? Consider both gases as ideal.
- A. ☐ The atoms in the helium sample will exhibit a narrower range of speeds than those in the argon sample.
 - B. ☐ The argon atoms exert a greater pressure on the walls of their container than the helium atoms.
 - C. ☐ The average speed of atoms in the helium sample will be greater than the average speed of atoms in the argon sample.
 - D. ☐ The volume of the helium sample is larger than the volume of the argon sample.
 - E. ☐ None of these statements is true.
5. **5 points.** Combustion of propane (C_3H_8) proceeds according to the balanced chemical equation below. In a combustion reaction occurring at constant temperature, the rate of production of CO_2 is 0.036 M/min. What is the corresponding rate of consumption of C_3H_8 ? Enter your response in M/min to the nearest 0.001 M/min *with appropriate sign for a rate of consumption*.



M/min

6. **6 points.** Fill in the blanks below to specify the orders of NO and ONOO^- in the rate law for the given reaction. Then, indicate the units of the rate constant in the third dropdown.



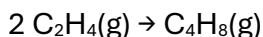
Experiment	$[\text{NO}_2]_i$ (M)	$[\text{ONOO}^-]_i$ (M)	Initial Rate (M/min)
1	0.00125	0.00125	2.03×10^{-6}
2	0.00125	0.000625	1.02×10^{-5}
3	0.000625	0.00250	2.03×10^{-6}
4	0.000625	0.00375	3.05×10^{-5}

Order of NO: [☐ 0 ☐ 1 ☐ 2]

Order of ONOO^- : [☐ 0 ☐ 1 ☐ 2]

Units of the rate constant k : [☐ M/min ☐ 1/min ☐ 1/(M·min)]

7. **5 points.** The measurements below were made during the reaction of ethylene (C_2H_4) with itself, a second-order reaction with $k = 0.032 \text{ M}^{-1} \text{ s}^{-1}$.

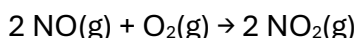


Determine the molarity of ethylene remaining at time $t = 300 \text{ s}$. Enter your response in moles per liter (M) to the nearest 0.001 M.

Time (s)	$[\text{C}_2\text{H}_4] \text{ (M)}$
0	0.100
100	0.076
200	0.061
300	?

mol/L

8. **5 points.** Oxidation of NO occurs according to the balanced chemical equation and mechanism below. Fill in the blanks to describe this mechanism.



1. $2 \text{NO} + \text{O}_2 \rightarrow (\text{ONO})_2$
2. $(\text{ONO})_2 \rightarrow 2 \text{NO}_2$

The species $(\text{ONO})_2$ is a

[☐ reactant ☐ product ☐ catalyst ☐ intermediate ☐ transition state].

The rate law for the reaction is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$. Based on this rate law, the slow step in the mechanism

[☐ is the first step. ☐ is the second step. ☐ cannot be determined.]

9. **5 points.** The reaction $\text{A}(\text{g}) + \text{B}(\text{g}) \rightarrow \text{C}(\text{g})$ has a rate law: $\text{rate} = k[\text{A}]$. If the concentration of B is halved, what is the effect on the rate of the reaction?
- A. ☐ The rate doubles.
 - B. ☐ The rate increases by a factor of four.
 - C. ☐ The rate decreases by half.
 - D. ☐ The rate decreases by a factor of four.
 - E. ☐ The rate does not change.

10. **5 points.** Consider two reactions **A** and **B** with equal activation energy, both occurring in a single elementary step. Reaction **A** has more productive orientations (i.e., orientations that lead to reaction upon collision) than reaction **B**. Which of the following statement(s) *must* be true?

- [☐ Must be true ☐ Could be false] Reaction **A** is catalyzed and reaction **B** is not.
[☐ Must be true ☐ Could be false] The frequency factor of reaction **A** is less than the frequency factor of reaction **B**.
[☐ Must be true ☐ Could be false] The overall order of reaction **A** is greater than the overall order of reaction **B**.
[☐ Must be true ☐ Could be false] The frequency factor of reaction **A** is equal to the frequency factor of reaction **B**.

11. **5 points.** Calcium carbonate is consumed in the presence of two equivalents of hydrochloric acid according to the balanced chemical equation below.



What is the correct equilibrium equation for this reaction?

- A. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]^2}$
B. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2][\text{H}_2\text{O}]}{[\text{CaCO}_3][\text{HCl}]^2}$
C. ☐ $K = \frac{[\text{HCl}]^2}{[\text{CaCl}_2][\text{CO}_2]}$
D. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]}$
E. ☐ $K = [\text{CaCl}_2][\text{CO}_2][\text{HCl}]^2$

12. **6 points.** Calcium carbonate decomposes according to the following balanced chemical equation. The equilibrium constant for this process is 1.8×10^7 .



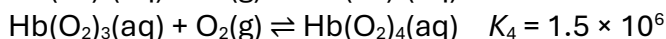
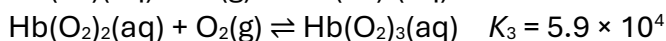
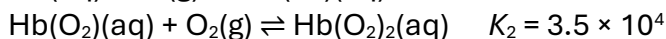
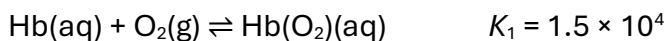
Consider a reaction vessel containing 0.200 M CO_2 , 0.500 M CaCl_2 , and 0.0500 M HCl . To determine the equilibrium concentrations of reactants and products in this system, what expressions should appear in the Change row of an ICE table? x is the number of reaction occurrences between the initial state and equilibrium. Enter "0" if the system is already in equilibrium.

HCl: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

CaCl_2 : [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

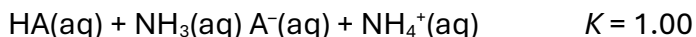
CO_2 : [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

13. **5 points.** Hemoglobin (Hb) is associated with four oxygen-binding equilibria.



What is the equilibrium constant for the reaction $\text{Hb}(\text{O}_2)_4(\text{aq}) \rightleftharpoons \text{Hb}(\text{aq}) + 4 \text{O}_2(\text{g})$?

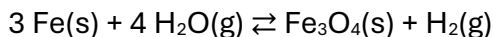
- A. ☐ 4.6×10^{19}
 B. ☐ 3.5×10^8
 C. ☐ 2.2×10^{-20}
 D. ☐ 1.1×10^{-5}
 E. ☐ 6.7×10^{-24}
14. **5 points.** The reaction of acetic acid (HA) with ammonia (NH_3) in aqueous solution is very close to perfectly reversible.



In a solution initially containing 0.400 M HA and 0.400 M NH_3 , what is the equilibrium concentration of acetate ion (A^-)? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.

mol/L

15. **6 points.** Consider a system in equilibrium with respect to the reaction below. Indicate whether each of the following interventions will cause the generation of more reactants ($\leftarrow$), more products ($\rightarrow$), or neither ($\rightleftharpoons$). The reaction is exothermic with $\Delta H^\circ = -151 \text{ kJ/mol}$.



[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Decreasing the reaction temperature

[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Removing $\text{H}_2\text{(g)}$ from the reaction vessel

[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Decreasing the volume of the reaction vessel

16. **5 points.** What is the correct chemical formula for the conjugate base of HCrO_4^- ?

- A. ☐ CrO_4^{2-}
- B. ☐ H_2CrO_4
- C. ☐ HCrO_4^{2-}
- D. ☐ HCrO_4
- E. ☐ HCrO_3^-

17. **5 points.** What is the pH of a solution made by dissolving $3.0 \times 10^{-3} \text{ mol Sr(OH)}_2$ in enough water to make 3.00 L of solution? Enter your response to the nearest 0.01.

18. **5 points.** Indicate whether each of the following is a state variable or path variable.

[☐ state variable ☐ path variable] Work

[☐ state variable ☐ path variable] Energy

[☐ state variable ☐ path variable] Volume

19. 5 points. A gas is contained in a cylinder capped with a movable piston. The external pressure on the piston is constant. The gas is heated, causing it to expand and push the piston outward. Which of the following statements is *true* concerning the work done by the gas during this process?

- A. ☐ The work done by the gas is negative because heat is added to the system.
- B. ☐ No work is done because the external pressure remains constant.
- C. ☐ The work done by the gas is zero because only the temperature changes.
- D. ☐ The work done on the gas is positive because the piston moves outward.
- E. ☐ The work done by the gas is positive because the system expands against the external pressure.